

MUHAMMAD BILAL AKRAM

Software Engineer | Full-Stack • ML/AI • Mobile Development

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PROFESSIONAL SUMMARY

Highly motivated Software Engineering graduate (COMSATS University, 2026 | CGPA: 3.48) with 2 industry internships and 10+ shipped projects spanning full-stack web, machine learning, and cross-platform mobile development. At Developerhub Corporation, single-handedly delivered 2 production web applications, a multi-vendor e-commerce platform and a real-time entrepreneur-investor network from zero to live deployment. At Arch Technologies, trained and deployed 4 ML models (spam classifier, digit recognizer, flower classifier, House price prediction model) with model explainability using LIME and SHAP. Final Year Project (VendoMart) integrates Gemini AI, real-time WebSockets, Stripe payments, and a full React Native web and mobile app across a 3-sided marketplace. Fluent across Python/Django, Node.js/Express, React Native, PostgreSQL, Scikit-learn, and TensorFlow and consistently productive at the intersection of all three. Targeting Junior Software Engineer, Backend Developer, ML Engineer, or React Native Developer roles.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C++, Kotlin, HTML5, CSS3, SQL
Frameworks	Django, React Native (Expo), Node.js, Express.js, React.js, Bootstrap, Tailwind CSS / NativeWind, Streamlit, Daphne, Socket.IO
ML / AI	Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Mediapipe, Hugging Face Transformers, SHAP, LIME
Databases	PostgreSQL, MySQL, SQLite3, MongoDB, MS SQL, Neon
Tools	Git, GitHub, Postman, Figma, JIRA, Draw.io, Lucidchart, JWT, Stripe API, Google Maps API, Cloudinary, Sequelize, Redux Toolkit
Concepts	REST APIs, WebSockets, Real-Time Systems, RBAC, MVC, System Design, API Design, Agile, CI/CD Fundamentals, OOP, Design Patterns

WORK EXPERIENCE

Full-Stack Software Development Intern | Developerhub Corporation

Jul 2025 – Aug 2025

- Engineered MarketVibe, a production-grade multi-vendor e-commerce platform serving 3 distinct user roles (admin, seller, buyer) with 8+ core modules, Stripe Connect payments, dynamic product catalog, smart search, gift campaigns, quote system, and order management deployed live on PythonAnywhere.
- Architected CoFound, a real-time professional networking platform connecting entrepreneurs and investors, featuring bidirectional WebSocket messaging (Django Channels + Daphne), meeting scheduling, fund-raising feeds, and JWT-secured RBAC reducing manual networking friction to a single platform.

Machine Learning Intern | Arch Technologies

Aug 2025 – Sep 2025

- Trained an Email Spam Classifier using Python and Scikit-learn with TF-IDF vectorization, achieving multi-model benchmarking across 3+ classifiers and integrating LIME explainability to surface per-prediction feature importance; packaged as an interactive Streamlit application.
- Engineered an MNIST Digit Recognition pipeline comparing Logistic Regression against a Convolutional Neural Network (CNN) in TensorFlow/Keras on 70,000 training samples, deploying a real-time Streamlit app that accepts live drawn-digit input for instant inference.
- Constructed an Iris Flower Classification system covering the full ML lifecycle across 3 species and 150+ samples — data preprocessing, feature engineering, cross-validated model selection, and rich SHAP + Plotly + Seaborn explainability dashboards.

KEY PROJECTS

VendoMart — Final Year Project — *React Native, Expo, TypeScript, Node.js, Express.js, PostgreSQL, Sequelize, Socket.IO, Redux Toolkit, NativeWind, Stripe, Cloudinary, Gemini AI, JWT*

- Spearheaded end-to-end development of a 3-sided AI-powered grocery marketplace (vendors, consumers, delivery riders) as a cross-platform React Native mobile app, unifying 4+ role-specific dashboards within a single codebase.
- Integrated Gemini AI for intelligent product recommendations, Socket.IO for real-time order tracking across 3 actor types, Stripe for secure payments, and Cloudinary for media management handling 5+ third-party API integrations simultaneously.
- Designed a normalized PostgreSQL schema with Sequelize ORM, JWT authentication, and RBAC; structured a RESTful Express.js backend with 20+ API endpoints supporting the full marketplace lifecycle.

Real-Time Sign Language Interpreter — *Python, Streamlit, OpenCV, Mediapipe, Scikit-learn, Hugging Face Transformers, Pyttsx3*

- Created a computer vision pipeline using Mediapipe to detect and classify 26 ASL hand-gesture landmarks in real time, feeding predictions into Hugging Face Transformers to generate contextual natural-language sentences with multi-language translation output.
- Delivered a fully interactive Streamlit application with text-to-speech (Pyttsx3), conversation history, and live webcam inference awarded faculty recognition for technical complexity and real-world accessibility impact.

CoFound — Entrepreneur-Investor Networking Platform (Live) — *Django, SQLite3, Daphne, WebSockets, Bootstrap, GitHub*

- Launched a full-featured professional networking platform from scratch to live deployment on PythonAnywhere, supporting 3 user roles (Admin / Entrepreneur / Investor) with profile management, meeting scheduling, and fund-raising content feeds.
- Wired real-time bidirectional messaging using Django Channels and Daphne (ASGI), enabling WebSocket-based chat with zero-refresh delivery — live at cofound.pythonanywhere.com.

MarketVibe — Multi-Vendor E-Commerce Platform (Live) — *Django, SQLite3, Stripe Connect, Bootstrap, JavaScript*

- Deployed a production multi-vendor marketplace on PythonAnywhere with Stripe Connect split-payments between platform and sellers, supporting 5+ seller management features including storefront creation, inventory control, and order fulfillment — live at marketvibe27.pythonanywhere.com.
- Incorporated advanced buyer-side features: smart keyword search, newsletter subscription, gift box campaign builder, and a quotes-request system enabling B2B-style product negotiations.

BUZTable — Restaurant Reservation & Food Ordering System — *Django, SQLite3, Stripe API, Bootstrap, JavaScript*

- Authored a complete restaurant management system covering 4 core workflows, table reservation, online food ordering, real-time inventory management, and customer reviews in a single Django application. Secured end-to-end payment flows using Stripe API with transaction validation, reducing payment friction to a single checkout step for restaurant patrons — live at buztable.pythonanywhere.com

Photo Gallery — Museum-Inspired Social Platform (Live) — *Django, Tailwind CSS, SQLite3, JavaScript*

- Crafted a museum-style photo-sharing platform with public/private visibility toggles, social interactions (likes and comments), and personal collection curation — live at photogallery.pythonanywhere.com.

EDUCATION

Bachelor of Software Engineering

SEP 2022 -JUNE 2026

COMSATS University Islamabad — Sahiwal Campus | CGPA: 3.48 / 4.00

CERTIFICATIONS

Python and Django Full-Stack Web Development Bootcamp — Udemy

Python for Data Science and Machine Learning Bootcamp — Udemy

Design Patterns Made Easy — Udemy

ACHIEVEMENTS & RECOGNITION

- Hardware-Software AI Project Competition — Recognised by COMSATS University for integrating voice control into a Bluetooth-controlled remote car using Arduino + Android; awarded for innovation in IoT and embedded systems at the university project showcase.
- Academic Project Recognition — BUZTable and Real-Time Sign Language Interpreter singled out by faculty during formal evaluations for practical real-world applications and exceptional technical complexity.